IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

**MATHEMATICS 1B**

Monday, 26 April 10:00 am

Time allowed: 2:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions. Question 1 & 2 are Mathematics, worth 30 & 45 marks, question 3 is Signals and Communications, worth 25 marks.**

*NO CALCULATORS ALLOWED.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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# THE QUESTIONS

1. a) We would like to evaluate the double integral

$$I = \int_0^1 \left( \int_x^{\sqrt{2-x^2}} f(x,y)\, dy \right) dx \, .$$

Begin by sketching the region of integration, and rewrite the integral in terms of polar coordinates:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta \, .$$

[You may assume the result $dx\, dy = r\, dr\, d\theta$.]

Now solve the integral for

$$f(x,y) = \sqrt{x^2+y^2} \left[ \cos^{-1}\left( \frac{x}{\sqrt{x^2+y^2}} \right) + \sin^{-1}\left( \frac{y}{\sqrt{x^2+y^2}} \right) \right] \, .$$

[ 7 ]

b) A function is defined as $f(x) = L - x$ on the interval $[0, L]$, where $L$ is a positive constant. We would like to represent this function on $[0, L]$ as a Fourier cosine series.

(i) Let $g(x)$ be the appropriate extension of $f(x)$ so that the Fourier series for $g(x)$ will consist of a constant and cosine terms. Sketch $g(x)$ for $-2L < x < 2L$. [ 2 ]

(ii) Obtain the cosine series for $g(x)$ in terms of $L$, writing down the first five terms. [ 5 ]

(iii) If the first cosine term of the series is $\frac{4}{\pi}\cos x$, deduce the value of $L$ and write out the series. [ 2 ]

(iv) Use Parseval's theorem and (i)-(iii) to find $\sum_{n=1}^{\infty} \frac{1}{n^4}$ . [ 4 ]

c) In your own words, explain:
(i) how to decide the existence of a function $F(x,y)$, so that a first-order Ordinary Differential Equation (ODE)

$$M(x,y)\frac{dy}{dx} + N(x,y) = 0$$

can be written in terms of the total differential $dF$, and
(ii) how a solution of the ODE is obtained by first finding $F(x,y)$. [ 5 ]

2. a) Solve the first-order ODE

$$\left(\frac{1}{2}t^3 e^{\frac{1}{2}x} - t\cos x + 2x\right)\frac{dx}{dt} = t^2 + \sin x - 3t^2 e^{\frac{1}{2}x},$$

subject to the initial condition $x(1) = 0$. [ 5 ]

b) (i) Find and classify the stationary points of the function

$$f(x,y) = (y-x)(y^2-x).$$ [ 5 ]

(ii) In your own words, explain why a saddle point lies at the intersection of two contour lines, and how this can be used to sketch the contours of a function of two variables. A sketch may be useful here! [ 5 ]

(iii) Sketch the contour where $f(x,y) = 0$ and then construct the remaining contouirs of $f(x,y)$, carefully explaining how you use (i) and (ii). [ 5 ]

c) Given the matrix

$$A = \begin{pmatrix} -2 & 1 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & \gamma \end{pmatrix},$$

find the value of $\gamma$ that makes the matrix singular. Taking $\gamma = -3$, use Gaussian elimination to find the inverse of $A$. [ 5 ]

3. a) (i) Give a sketch of the function

$$f(t) = \begin{cases} 0 & , \quad 0 < t \leq a \, , \\ t - a & , \quad a < t \leq 2a \, , \\ a & , \quad 2a < t \, . \end{cases}$$

where $a > 0$. Show that the Laplace transform of $f(t)$ is
$\bar{f}(s) \; = \; \frac{1}{s^2} \left( e^{-as} - e^{-2as} \right)$ . [ 5 ]

(ii) Given the second-order Ordinary Differential Equation (ODE)

$$\frac{d^2x}{dt^2} + x = \begin{cases} 0 & , \quad 0 < t \leq \pi \, , \\ t - \pi & , \quad \pi < t \leq 2\pi \, , \\ \pi & , \quad 2\pi < t \, . \end{cases}$$

which satisfies $x(0) = x'(0) = 0$, use (i) and Laplace Transforms to solve the ODE.

Give the solution in the form

$$x(t) = \begin{cases} x_1 & , \quad 0 < t \leq \pi \, , \\ x_2 & , \quad \pi < t \leq 2\pi \, , \\ x_3 & , \quad 2\pi < t \, , \end{cases}$$

simplifying as much as possible. Take care to explain each step. [ 10 ]

b) Given the matrix

$$A = \begin{pmatrix} 2 & -3 & -1 \\ -2 & 3 & 2 \\ 1 & -2 & 3 \end{pmatrix} ;$$

(i) Using two row operations, explain in words why $A$ cannot be written as the product $A = LU$ where $L$ is lower triangular and $U$ upper triangular. Hence, identify the permutation matrix $P$ needed so that $PA$ can be written as $LU$. [ 5 ]

(ii) Use Gaussian elimination to transform $PA$ into the upper triangular $U$ and write down the elimination matrices needed for this. Write down the inverses of the elimination matrices, and hence find $L$, so that $PA = LU$. [ 5 ]

4. a) You are given four points $A(-1,1,5)$, $B(-3,1,1)$, $C(1,1,0)$ and $D(2,-1,-2)$, with two lines, $L_1$ through $A$ and $B$, and $L_2$ through $C$ and $D$.

(i) Find the vector equations of $L_1$ and $L_2$, in terms of four vectors and two parameters, $\lambda, \mu \in \mathbb{R}$. Explain in your own words how to find points $P_1$ on $L_1$ and $P_2$ on $L_2$, so that $|\overrightarrow{P_1P_2}|$ is the shortest distance between the two lines. [ 5 ]

(ii) Write down the three equations in $\lambda, \mu$ and one further unknown, which you need to find $P_1$ and $P_2$. Write these equations in matrix form $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$. Is $A$ invertible or singular, and how do we know this without any calculations? [ 4 ]

(iii) Use Gaussian elimination to obtain the echelon form of the augmented coefficient matrix $(A : \underline{\mathbf{b}})$, and hence find $\underline{\mathbf{x}}$. Hence, obtain $P_1, P_2$ and the shortest distance. Finally, consider the situation where this question is badly posed: the lines intersect and the minimum distance is zero. Explain how $A$ would change in this case. [ 4 ]

b) Use Gaussian elimination to find the reduced echelon form of

$$A = \begin{pmatrix} 3 & 4 & 2 & -1 \\ 2 & 6 & -7 & 11 \\ 1 & 2 & -1 & 2 \end{pmatrix}.$$

Identify rank, pivot columns and free columns. Obtain bases for the four fundamental subspaces of $A$: column space, row space, nullspace and left nullspace, taking to care to explain how you obtain these. Confirm the relationships between the dimensions of the subspaces and the number of rows and columns of $A$. [ 12 ]